

Database Technologies – Assignment 10

PL-SQL

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, $\text{Perimeter} = 2 * (L + W)$ and $\text{Area} = L * W$.)

```
DELIMITER $$
```

```
CREATE PROCEDURE rectangle()
```

```
BEGIN
```

```
DECLARE L DECIMAL(10,2) DEFAULT 10;
```

```
DECLARE W DECIMAL (10,2) DEFAULT 5;
```

```
DECLARE perimeter INT;
```

```
DECLARE area INT;
```

```
SET perimeter = 2 * (L + W);
```

```
SET area = L * W;
```

```
SELECT L AS Length,
```

```
W AS Width,
```

```
perimeter AS Perimeter,
```

```
area AS Area;
```

```
END $$
```

```
DELIMITER ;
```

```
CALL rectangle();
```

```

W2_99546_Pratiksha>use classwork_db;
Database changed
W2_99546_Pratiksha>DELIMITER $$
W2_99546_Pratiksha>CREATE PROCEDURE rectangle()
-> BEGIN
-> DECLARE L DECIMAL(10,2) DEFAULT 10;
-> DECLARE W DECIMAL (10,2) DEFAULT 5;
-> DECLARE perimeter INT;
-> DECLARE area INT;
-> SET perimeter = 2 * (L + W);
-> SET area = L * W;
-> SELECT L AS Length,
-> W AS Width,
-> perimeter AS Perimeter,
-> area AS Area;
-> END $$
Query OK, 0 rows affected (0.04 sec)

W2_99546_Pratiksha>DELIMITER ;
W2_99546_Pratiksha>CALL rectangle();
+-----+-----+-----+-----+
| Length | Width | Perimeter | Area |
+-----+-----+-----+-----+
| 10.00 | 5.00 | 30 | 50 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```

DELIMITER//
CREATE PROCEDURE number_calculation()
BEGIN
DECLARE num INT DEFAULT 5;
INSERT INTO temp
VALUES (num, num * num, num * num * num);
END//

```

```

W2_99546_Pratiksha>CREATE TABLE temp(num INT,square INT,Cub INT);
Query OK, 0 rows affected (0.03 sec)

W2_99546_Pratiksha>DELIMITER //
W2_99546_Pratiksha>CREATE PROCEDURE number_calculation()
    -> BEGIN
    -> DECLARE num INT DEFAULT 5;
    -> INSERT INTO temp
    -> VALUES (num, num * num, num * num * num);
    -> END //
Query OK, 0 rows affected (0.01 sec)

W2_99546_Pratiksha>DELIMITER ;
W2_99546_Pratiksha>Call number_calculation();
Query OK, 1 row affected (0.01 sec)

W2_99546_Pratiksha>SELECT * FROM temp;
+-----+-----+-----+
| num  | square | Cub  |
+-----+-----+-----+
| 5    | 25     | 125  |
+-----+-----+-----+
1 row in set (0.00 sec)

W2_99546_Pratiksha>DELIMITER ##

```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are $C = (F - 32) * 5 / 9$ $F = 9 / 5 * C + 32$
DELIMITER ##

```

CREATE PROCEDURE convert_temp(IN temp DECIMAL(10,2), IN choice CHAR(1))
BEGIN
    IF choice = 'F' THEN
        SELECT (temp - 32) * 5 / 9 AS Celsius;
    ELSEIF choice = 'C' THEN
        SELECT (9 * temp / 5) + 32 AS Fahrenheit;
    ELSE
        SELECT 'Invalid choice' AS Message;
    END IF;
END ##
DELIMITER ;
CALL convert_temp(100, 'F');
CALL convert_temp(37, 'C');

```

```

W2_99546_Pratiksha>DELIMITER ##
W2_99546_Pratiksha>CREATE PROCEDURE convert_temp(IN temp DECIMAL(10,2), IN c
hoice CHAR(1))
-> BEGIN
-> IF choice = 'F' THEN
-> SELECT (temp - 32) * 5 / 9 AS Celsius;
-> ELSEIF choice = 'C' THEN
-> SELECT (9 * temp / 5) + 32 AS Fahrenheit;
-> ELSE
-> SELECT 'Invalid choice' AS Message;
-> END IF;
-> END ##
Query OK, 0 rows affected (0.01 sec)

W2_99546_Pratiksha>DELIMITER ;
W2_99546_Pratiksha>CALL convert_temp(100, 'F');
+-----+
| Celsius |
+-----+
| 37.777778 |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

W2_99546_Pratiksha>CALL convert_temp(37, 'C');
+-----+
| Fahrenheit |
+-----+
| 98.600000 |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```

DELIMITER //
CREATE PROCEDURE convert_inches(IN total_inches INT)
BEGIN
DECLARE yards INT;
DECLARE feet INT;
DECLARE inches INT;
SET yards = total_inches DIV 36;
SET feet = (total_inches MOD 36) DIV 12;
SET inches = total_inches MOD 12;
SELECT yards AS Yards,
feet AS Feet,
inches AS Inches;
END //
DELIMITER ;
CALL convert_inches(124);

```

```

W2_99546_Pratiksha>DELIMITER //
W2_99546_Pratiksha>CREATE PROCEDURE convert_inches(IN total_inches INT)
-> BEGIN
-> DECLARE yards INT;
-> DECLARE feet INT;
-> DECLARE inches INT;
-> SET yards = total_inches DIV 36;
-> SET feet = (total_inches MOD 36) DIV 12;
-> SET inches = total_inches MOD 12;
-> SELECT yards AS Yards,
-> feet AS Feet,
-> inches AS Inches;
-> END //

```

Query OK, 0 rows affected (0.01 sec)

```

W2_99546_Pratiksha>DELIMITER ;
W2_99546_Pratiksha>CALL convert_inches(124);

```

Yards	Feet	Inches
3	1	4

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

4. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```
DELIMITER //
```

```
CREATE PROCEDURE check_product(IN num1 DECIMAL(10,2), IN num2
DECIMAL(10,2))
```

```
BEGIN
```

```
IF num1 * num2 >= 100 THEN
```

```
SELECT 'Product is equal to or greater than 100' AS Result;
```

```
ELSE
```

```
SELECT 'Product is less than 100' AS Result;
```

```
END IF;
```

```
END //
```

```
DELIMITER ;
```

```
CALL check_product(10, 12);
```

```

W2_99546_Pratiksha>DELIMITER //
W2_99546_Pratiksha>CREATE PROCEDURE check_product(IN num1 DECIMAL(10,2), IN
num2
-> DECIMAL(10,2))
-> BEGIN
-> IF num1 * num2 >= 100 THEN
-> SELECT 'Product is equal to or greater than 100' AS Result;
-> ELSE
-> SELECT 'Product is less than 100' AS Result;
-> END IF;
-> END //

```

Query OK, 0 rows affected (0.01 sec)

```

W2_99546_Pratiksha>DELIMITER ;
W2_99546_Pratiksha>CALL check_product(10, 12);

```

Result
Product is equal to or greater than 100

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)